Venkata Sandeep Kumar Reddy

DATA ENGINEER | CLOUD DATA | STREAMING | ANALYTICS ENGINEERING

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Data Engineer with 3 years of professional experience building and supporting SQL/Python data workflows, alongside hands-on projects in real-time streaming, CDC/lakehouse engineering, and analytics platforms. Strengths include ETL/ELT, data quality, distributed processing, dimensional modeling, and reliable data delivery.

TECHNICAL SKILLS

Programming: Python, SQL, PySpark

Data pipelines: ETL/ELT, CDC, incremental processing, deduplication, idempotency, replay, data quality, reconciliation

Streaming & compute: Apache Kafka, Apache Spark, Spark Streaming, batch and distributed processing

Storage & modeling: PostgreSQL, Parquet, Apache Iceberg, lakehouse/warehouse, dimensional modeling, SCD, data marts

Cloud & platform: AWS S3, Glue Data Catalog, Athena, IAM, KMS, CloudWatch, Terraform, Airflow, Docker, Kubernetes, GitHub Actions, CI/CD

APIs & analytics: FastAPI, React, Streamlit, scikit-learn, Isolation Forest, testing, observability, performance benchmarking

PROFESSIONAL EXPERIENCE

Data Engineer | Mar Symantech Private Limited | September 2023 – Present

- Built, maintained, and automated ETL/ELT pipelines using Python and SQL to load structured and semi-structured production data into cloud databases and warehouses.
- Applied data-cleaning, schema-validation, and deduplication checks to help prevent inconsistent or corrupted records from reaching downstream analytics.
- Scheduled and monitored batch jobs using Airflow and cron; investigated pipeline failures and resolved production data-processing issues.
- Improved SQL queries and transformations using indexing and execution-plan analysis to address processing bottlenecks and query response times.
- Prepared analytics-ready data marts and semantic layers; documented data definitions and lineage to support reliable reporting.

SELECTED ENGINEERING PROJECTS

Enterprise AI-Powered Real-Time EV Fleet Data Platform

Python • Kafka • Spark • PostgreSQL • FastAPI • Docker • Kubernetes

Processed 98,061,214 telemetry records (98,060,402 valid; 812 rejected), generating 160,774 feature windows and 8,251 AI samples. Isolation Forest F1: 0.8619; API benchmark: 20/20 successful requests, 77.278 ms p95.

Cloud-Native Lakehouse, CDC & AI-Ready Data Platform

PostgreSQL pgoutput CDC • Raw/Bronze/Silver/Gold

Implemented and tested INSERT/UPDATE/DELETE CDC with LSN progression, ordering, deduplication, idempotency, and checkpoints. Validation: 15/15 automated tests, 63/63 data-quality checks, 60/60 reconciliation checks; 9 Raw, 9 Bronze, and 6 Gold datasets/products.

RetailIQ — Modern Data Warehouse & Analytics Engineering Platform

PostgreSQL ELT • Dimensional modeling • SCD • Data marts • Streamlit

Built incremental analytical models and BI-ready products with facts/dimensions, SCD, data marts, and semantic views. Validated 34/34 data-quality checks and 9/9 integration tests; delivered a six-page Streamlit dashboard.

EDUCATION

B.Tech, Data Science — Presidency University (2019–2023) | PGP, Data Science and Machine Learning — Intellipaat (2024)